

4. Synchronous FIFO:

> Reference model:

• Data storage: Set of data mimicking the actual data.
(data = golden queue [57])

→ on valid wr & write request, data is pushed to back of queue. (push-back())

→ on valid read request, data is popped from front.
(pop-front())

→ full & empty flags are calculated on the internal queue
O & N depth.

> Test cases:

→ Reset

→ Write & Read

→ Fifo - full

→ Fifo - empty

→ Simultaneous Read & write.

→ Read & write when fifo is full.

→ Read & write when fifo is empty.

> Coverage:

bins for i) wr → 0 & 1

ii) rd → 0 & 1

iii) status → full & empty.

iv) boundaries → all 0 & all 1.

v) cross coverage (wr x rd x status).